Technical Exposition

Data Preprocessing

Datasets Used: *sessions.csv*, *transactions.csv*, and *product_categories.csv*

We began with separate preprocessing (data cleaning) of each dataset using Pandas for its flexibility and speed. Due to file sizes, we used Google Colab for light processing such as preliminary EDA and transitioned to local environments for heavy operations, which required more memory, such as merging datasets and model generation.

Data cleaning (both on sessions.csv and transactions.csv)

For both datasets, we applied a 'domain_name' == ebay.com filter, as that's our targeted domain. On the event_date and event_time columns, we cast them to Pandas's datetime format and combined them into a single date-time column. For categorical values with -88, 99, and 00 (unknown and null), we converted them to NaN because unknown categories are essentially null.

sessions.csv

Dealing with null values

After data exploration using .info() and .describe(), we noticed various columns contained null values. For columns with null percentages that are less than 1%, we decide to drop the null rows due to the minimal loss of data. On the other hand, for columns with greater than 1% null values, we cleaned the data as follows:

1. **Ref_domain_name:** This column is 75.8% null. The null values indicated that there were no references to the session. Therefore, we one-hot-encoded the top 10 domains, keep other domains as "Other," and change null values to "No referral"
2. **Hoh_most_education:** most values are null or unknown (60%), so we decided to drop the column to prevent skewness and imputing more data than the known values
3. **Household_size, connection_speed, zip_code:** these columns exhibit a low null percentages (1.53, 7.00, and 2.83, respectively). We grouped each row by **census_region** and used their respective median and mode for a finer imputed value, as we believed the location of each session could influence other factors.

Feature engineering

The new features we created are `session_datetime`, `is_weekend`, and `is_weekday`, all derived from the datetime column that we created from the event_date. Session_datetime is simply the combination of the event_date and event_time columns in the Pandas datetime type. Is_weekend and is_weekday are binary columns that indicate the type of day of the event.

transactions.csv

Dealing with null values and dropping columns
For the transactions dataframe, we performed a similar process to the sessions dataset in dropping rows with null values. We also noticed the existence of the 'total_transactions_[YYYY]' columns, which were not represented in the given data dictionary. As we are skeptical of using them, we decided to drop those columns. Lastly, we dropped the product name, as we could simply use the information given prod_category_id and the product_categories table.

Merging Dataframe

Merging transactions with sessions
We found that the site_session_id on the transactions table is faulty, where a large portion of the table (year 2022-2024) would contain 0. Therefore, to merge with the sessions table, we investigated other candidate columns to merge on, such as the combination of machine_id and event date-time, to create a sessions table instead of merging on site_session_id. However, we realized that machine_id acts as a good proxy for unique users from our histograms below. Thus, we decided to merge on the machine_id column.

After merging, we dropped redundant columns, while maintaining the user demographics, session date-time, and product id.

Merging the merged dataframe with categories
We would love to understand what the product ID signifies, as that would help with understanding user purchases in deeper detail, helping us with user classifications.

Right here, we merged the dataframe on prod_category_id from the previously merged dataframe on the Product ID on the product_categories csv. We then drop the product ID, product categories, leaving only the item categories.

As clustering and logistic regression needs numerical input, we performed one hot encoding on the item categories, expanding into 37 new columns.

Merged Dataframe
After more cleaning and dropping columns, the merged dataframe consists of these columns as follows:

Column origin	Features
transactions	'machine_id', 'prod_category_id', 'domain_id_x', 'prod_qty', 'prod_totprice', 'basket_tot', 'transaction_hour', 'transaction_dayofweek', 'transaction_month'
sessions	'event_date', 'site_session_id', 'user_session_id', 'domain_id_y', 'pages_viewed', 'duration', 'census_region', 'household_size', 'hoh_oldest_age', 'household_income', 'children', 'racial_background', 'connection_speed', 'hispanic', 'zip_code', 'ref_domain_x' (where x is the name of the domain), 'is_weekend', 'is_weekday'
Product categories	'Item_cat_x' (where x is the item categories)

Customer Dataset

As our analysis is on classifying users based on their behavior, the grouped dataframe after merging is grouped by **machine_id**, which is representative of each user.

Columns

- **Demographics data:** as each user is registered with one demographic, we extract the first value we find. These columns such as household size, racial background, census region, and number of children.
- **Transactions/sessions data:** these data are different in each session. Therefore, we try to find meaning in it. Examples are:
    - Sum: basket total price, product total price, product quantity, number of weekday sessions, number of weekend sessions
    - Mean over sessions: basket price, product price, product quantity
- **Item category data:** we summed up all items in each category to see how many purchases a person made per item category.
- **New features for analysis** such as the sum of successful transactions, unsuccessful transactions, conversion rate (percentage of successful transactions), and average time spent per page viewed

Sessions Dataset

To predict whether a transaction is successful, we need to group the merged dataframe based on the **site_session_id**, obtained from the sessions.csv part (as the transactions' session_id is faulty). The columns are identical to the customer dataset, but having multiple machine_id rows, and having another column called is_transacted, showing whether that session led to a successful transaction ot not.

Observed Difficulties

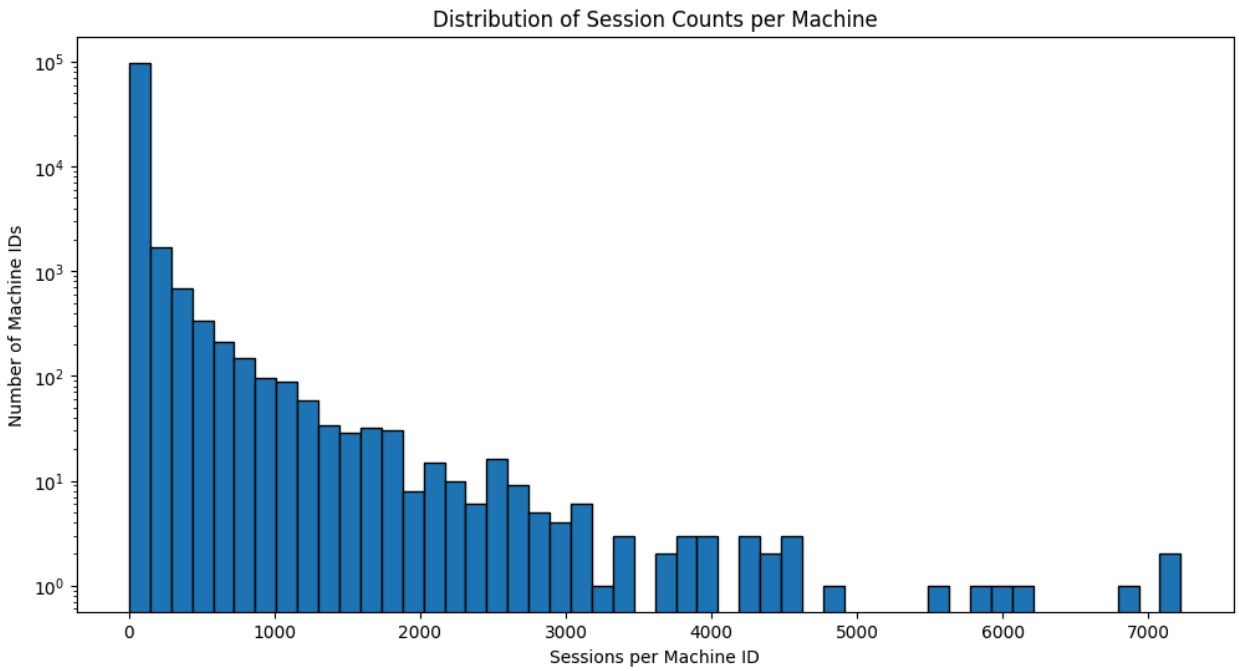
1. **Large file processing:** With Google Colab, processing and merging large files (especially sessions.csv) can cause the session to crash repeatedly. One way that we solved this problem was to work on a local server and then later export the cleaned merged dataset back for further analysis.
2. **Null and faulty data values:** After conducting exploratory analysis on the dataset, we noticed many null and faulty data entries as outlined above. We were able to circumvent this issue with clever imputations and column droppings.
3. **Number of item_categories columns:** This is for the clustering. When we selected only the top 5 item categories and classified other categories as ‘others’, the data is not detailed enough to know what the users are buying. However, if we use all of the item categories, the model becomes skewed with some random item categories that normal people might not buy. Therefore, we had to extract and try on different numbers of item categories and ended up with 10.

EDAs

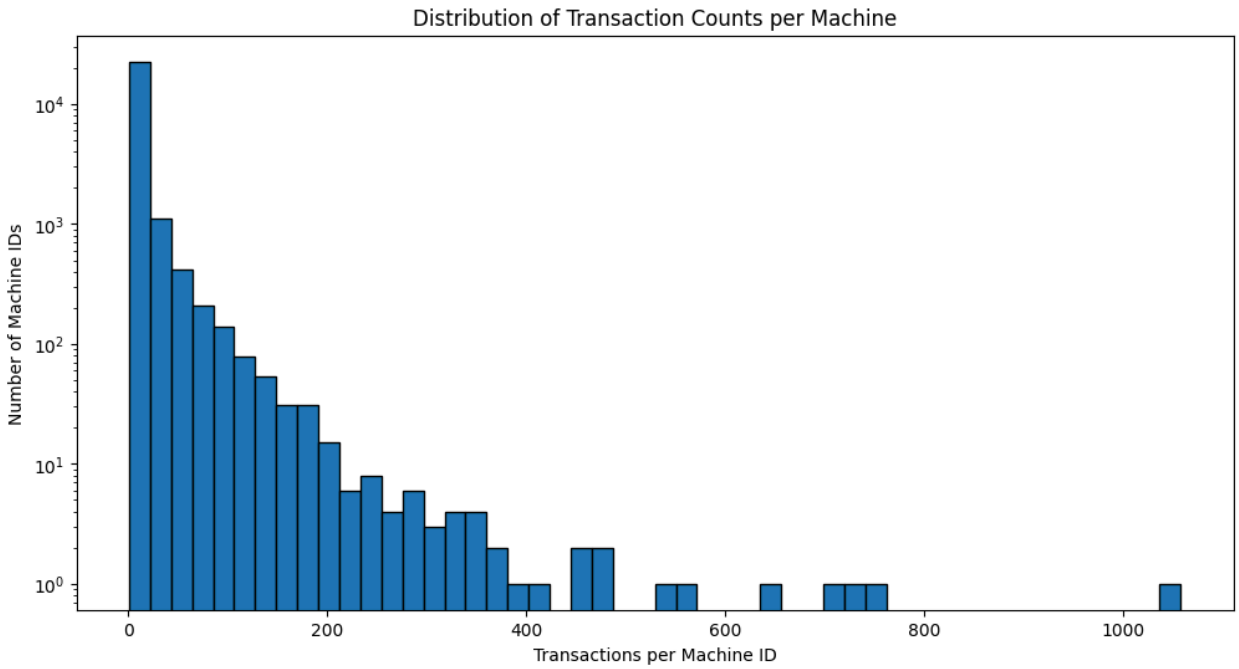
General Statistics

In this dataset, after filtering only eBay and grouping the merged dataframe based on customers(machine_id), transactions, and sessions, we found out that there are:

- 101,426 unique machine_ids (approximated as the number of unique customers), where each machine_id has approximately 30 session visits on average
- 147,947 unique successful transactions
- 2,771,371 unique sessions

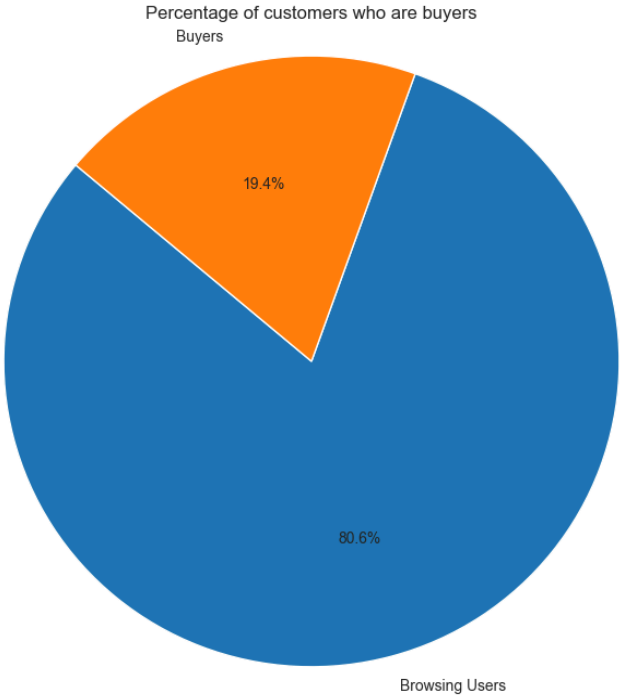


This graph shows the distribution of session counts per machine. The y-axis of the graph is a log scale. As we can see, the session counts per machine is quite skewed, with the vast majority of machine IDs having fewer than 200 sessions. As this is a log scale, around 1% of people are heavy users.



Based on the two graphs above, we see that most people make fewer than 50 transactions. **Most machine IDs (customers) have very few purchases**, which is confirmed by the buyers percentage below.

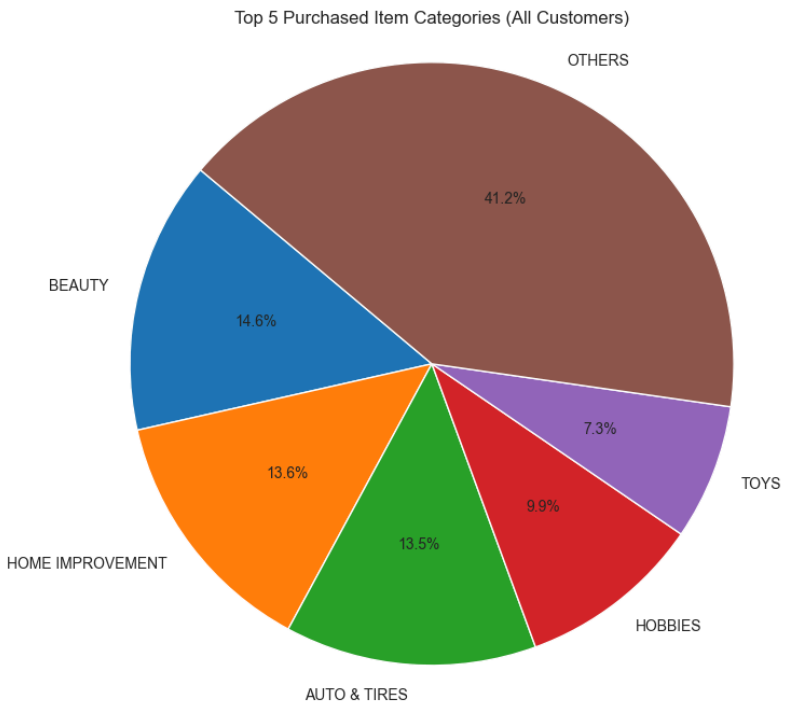
Out of all customers, there are only 19,635 users that have transacted successfully, accounting for **19.6% of all customers.**



Categories

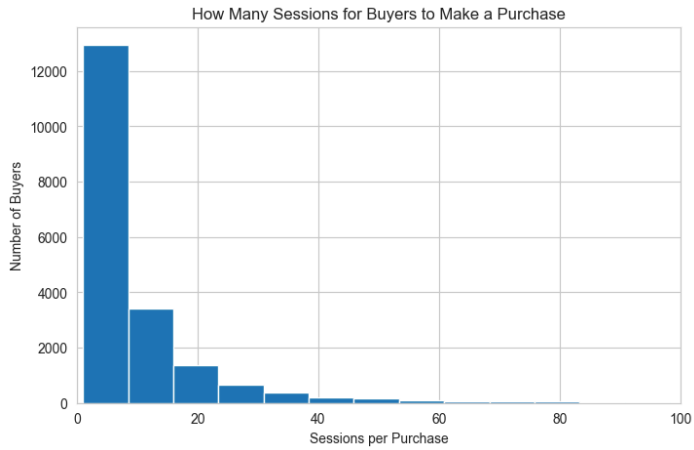
There are 12 reported categories and 50 item categories in total, 10 and 37 of which are on eBay, respectively.

The top 5 item categories for buyers are Beauty, Home Improvement, Auto & Tires, Hobbies, and Toys.



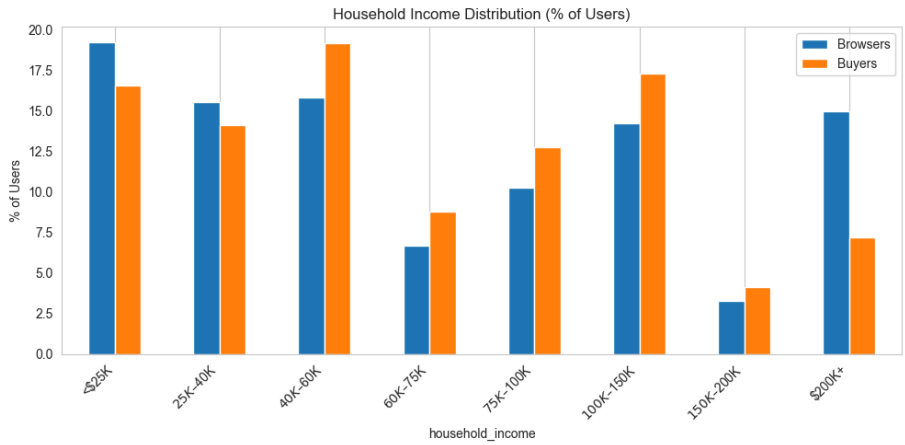
Buyers

- On average, buyers browse **11.62 sessions per purchase**. Here's the distribution of the number of sessions that buyers made to make a purchase:



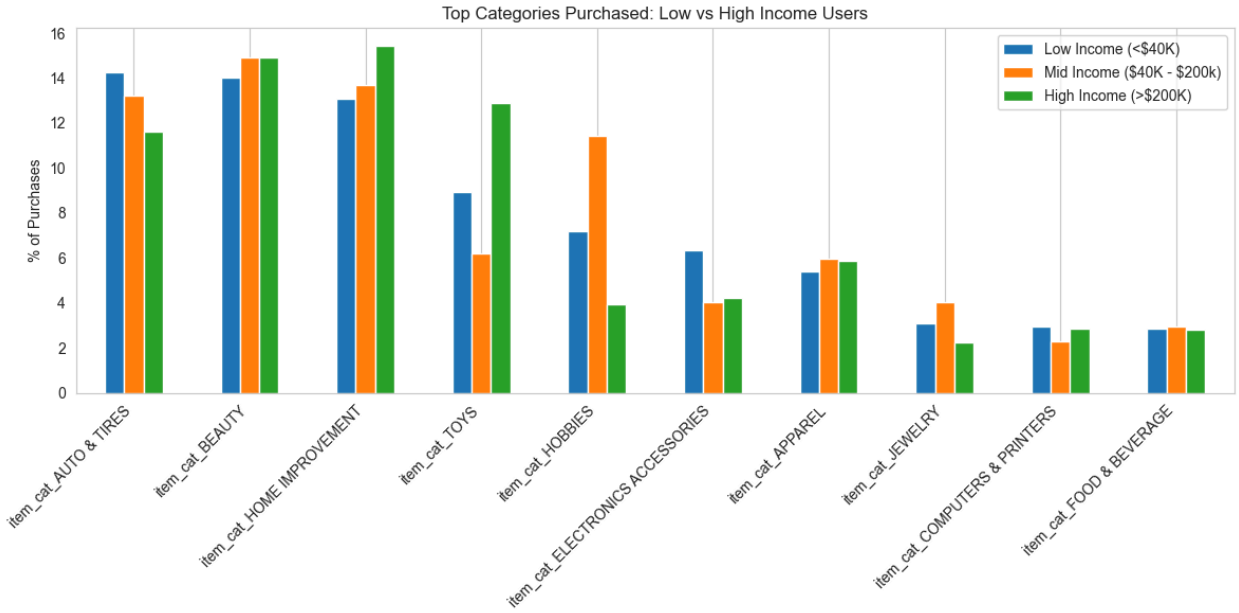
Buyer Demographics

1. Household income



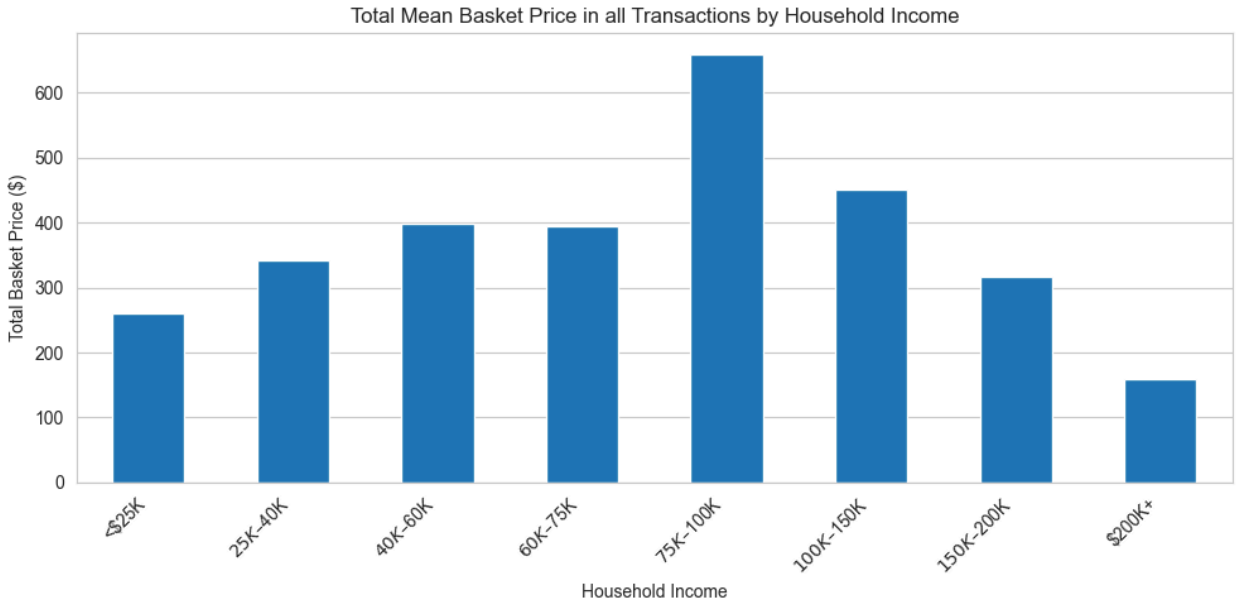
We noticed that household income impacts whether the person would buy things from eBay. Right here, it shows that the lower-end income (<$40k), and interestingly, the people with high-end incomes (>$200k) are window shoppers. We hypothesized that the product categories that these people shopped were different, so we decided to investigate further.

Further data digging into the users with lower incomes (<$40k), mid incomes ($40k-$200k) and high incomes (>$200k) shows different behaviors in product purchases:

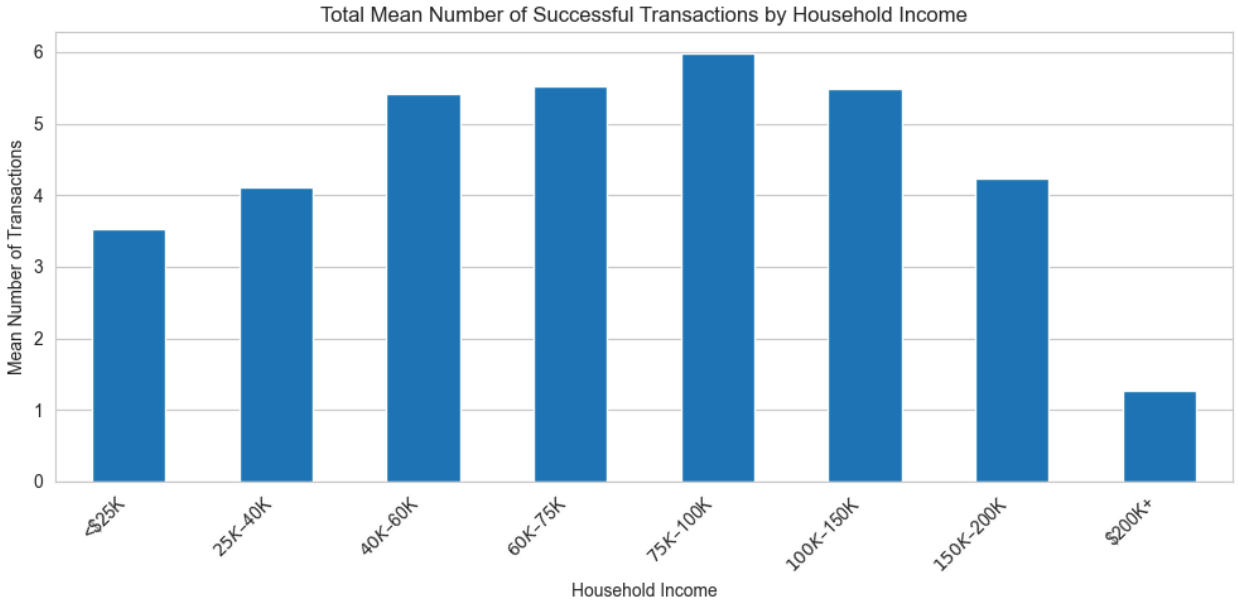


- Home improvement, beauty, and auto & tires are top categories for all segments.
- Higher-income people favor toys and home improvement more than other groups.

- The low-income group favors budget electronics more than other groups.
- The mid-income groups value hobbies more than other groups, and favor toys the least.
- Interestingly, high-income groups are least likely to buy jewelry many times on eBay, which might be because eBay is not the platform that they are looking for.

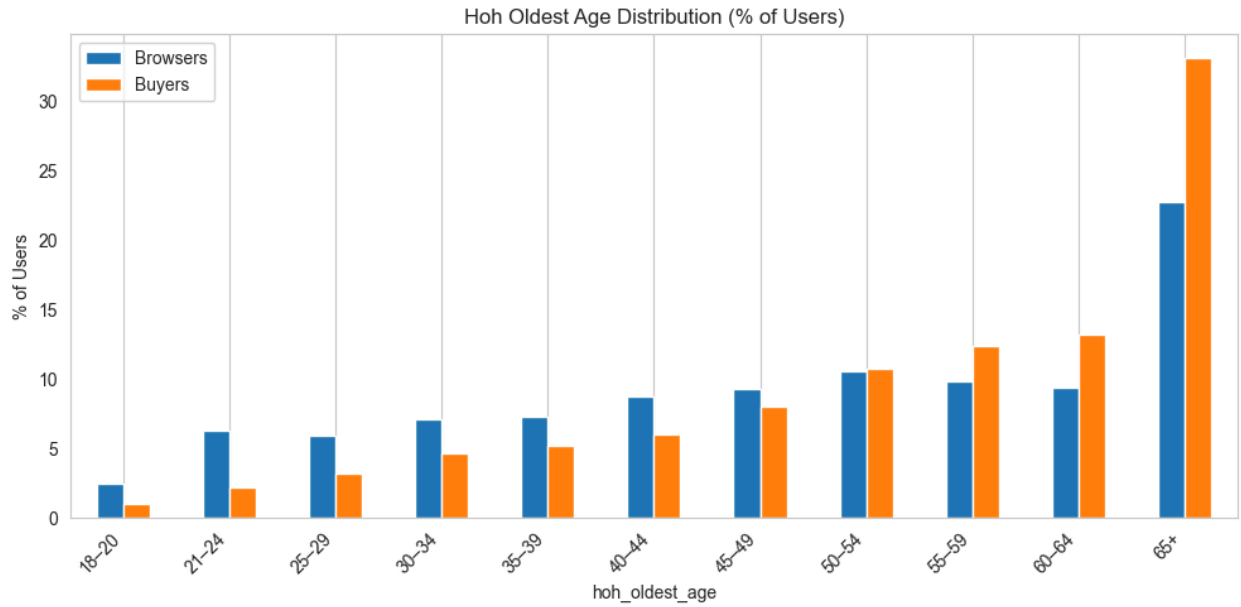


Household income also impacts how much a person buys from the platform as well: the most cumulative purchase is from the group with a household income of $75k - $100k.



Lastly, grouping by household income shows us that our main customers who repurchase the most are people with mid-income ($40k - $150k). High-income families aren't really our target, as the mean number of transactions is only around 1 (most of them aren't coming back).

2. Household Oldest Age

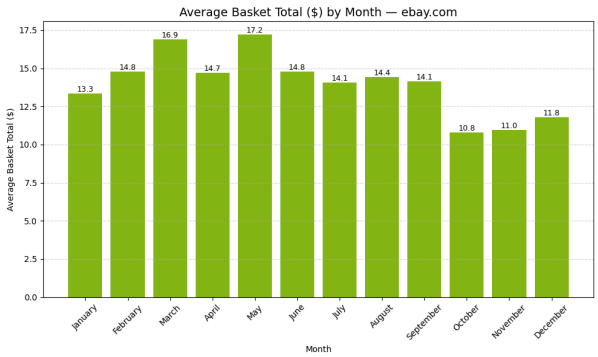


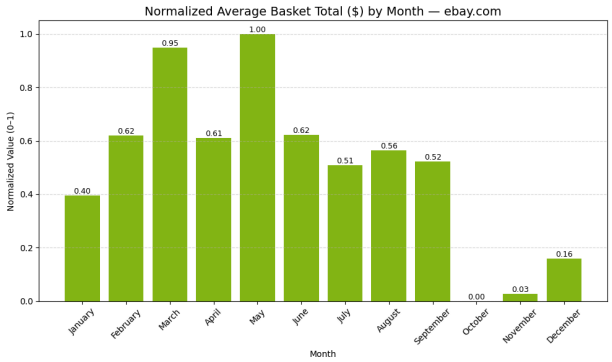
Households whose oldest person is young (<49 years old) tend to window shop more than households consisting of older people.

Purchase Sessions

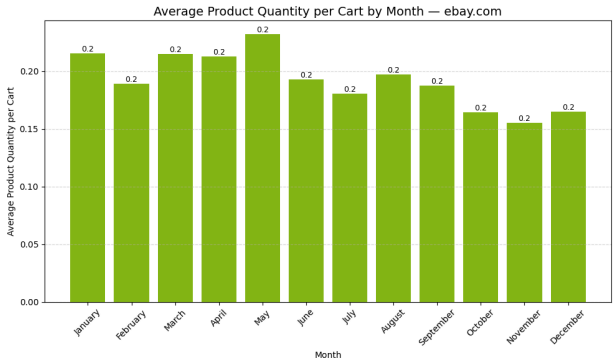
1. Seasonality

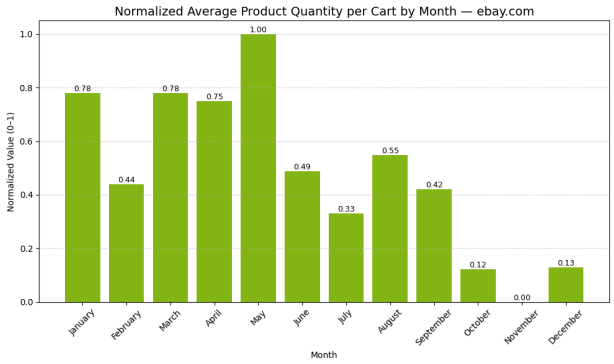
Seasonality affects spending, basket size, and the basket amount.





The graph above shows the average price of the purchased basket in each month. We noticed that **there are months with higher basket prices, which are March and May**, where we hypothesized that it might be the 'Spring Cleaning' effect, where people tend to clear out their homes and sell their unused items.





Similarly to the basket price, the **number of products in the basket is also the highest in May,** with approximately 6.2 items per purchase, implying in addition to high transaction count, the value of the items are also higher.

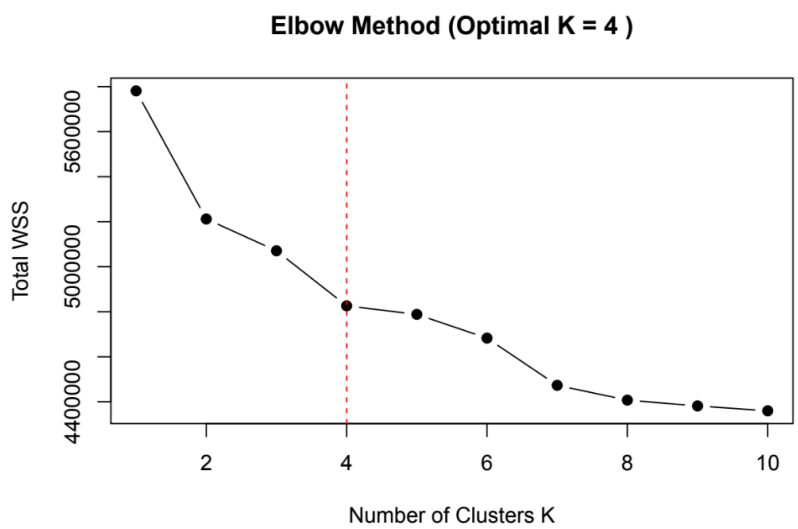
Modeling

K-means Clustering

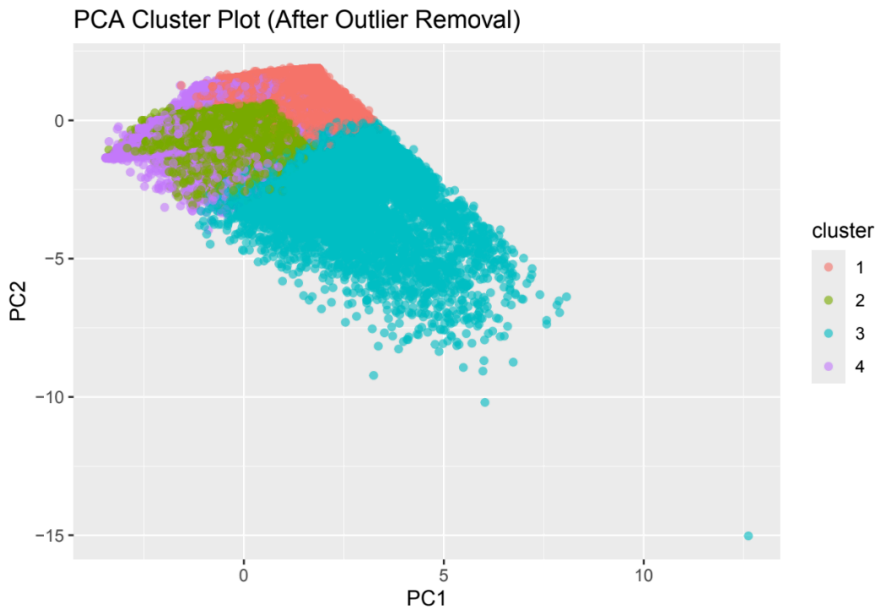
We used the K-means model because of its computational efficiency with a large number of customers and their spending habits that we found from the transaction and session data to find the ideal number of clusters, or the main customer segments on eBay.

Our steps are as follows:

1. **Elbow plot:** We plotted the number of clusters and the mean variance to find the optimal number of clusters; the targeted number of clusters would lie at the 'elbow' of the graph. The result shows that the optimal number of clusters (k) is 4, as shown in the figure below.



2. **K-means clustering simulation:** After our initial run of K-means, we saw that there were outliers in the data and skew in the data. To handle this, we removed outliers based on the principal components 1 and 2 to ensure that our clusters would be more interpretable.



Logistic Regression

After k-means clustering, we perform logistic regression on each of our customer partitions to reveal their respective behaviors and influential features, hoping to classify the customers into four personas for each cluster.

We first train the data using the session data and the outcome variable is_transacted, which is a 0/1 variable.

To avoid bias in the model, we used downsampling in the training set to ensure that there was equal representation in both transactional and non-transacting sections. This was done to ensure that the model could detect patterns in real-life purchase behavior and wouldn't be biased towards the large number of "non-transacting" sessions.

We applied all subset selection using BIC as the criterion to find the most predictive features within each cluster. We made sure to include session statistics as well as user demographic information and product categories.

Each cluster had its own separate logistic regression model using the most optimal features for each cluster to understand how each cluster's performance varies.

We used a ROC analysis to identify the optimal threshold that balances sensitivity and specificity. Predictions by the model were then classified into 0 or 1 based on this threshold, and model performance was evaluated using accuracy, sensitivity, and specificity.

Results

We followed a standardized model generation procedure for all of the clusters, revealing insightful data regarding impactful features. With each cluster, we analyzed the various features influencing customers' behavior. Moreover, we will primarily focus on features that are statistically significance with a Pr(>|z|), the p-value, less than 0.05.

In all of the clusters, the item categories are less indicative for this customer group according to their respective model and feature p-values, implying that what a customer buys does not majorly influence whether they will complete a transaction. This conclusion makes sense. Logically, to collect item category data, the customer must have already completed the transaction, meaning we cannot go backwards to predict the effects on customer likelihood to buy an item.

For each feature, we also calculated its influence on the odds of a transaction occurring. We derived the percentages using the Odds Ratio, which is simply e^(estimate). With the Odds Ratio, we are able to calculate percent change odds by using the following equation:

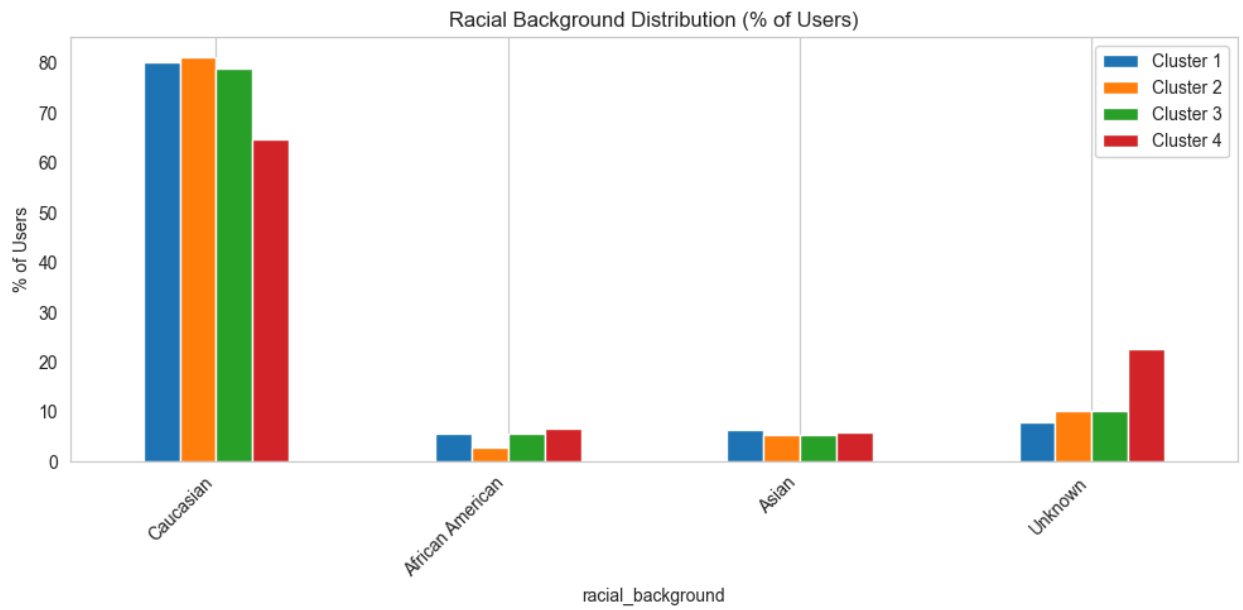
Percent Change in Odds = $(e^{\beta} - 1) \times 100$

K-means clustering results

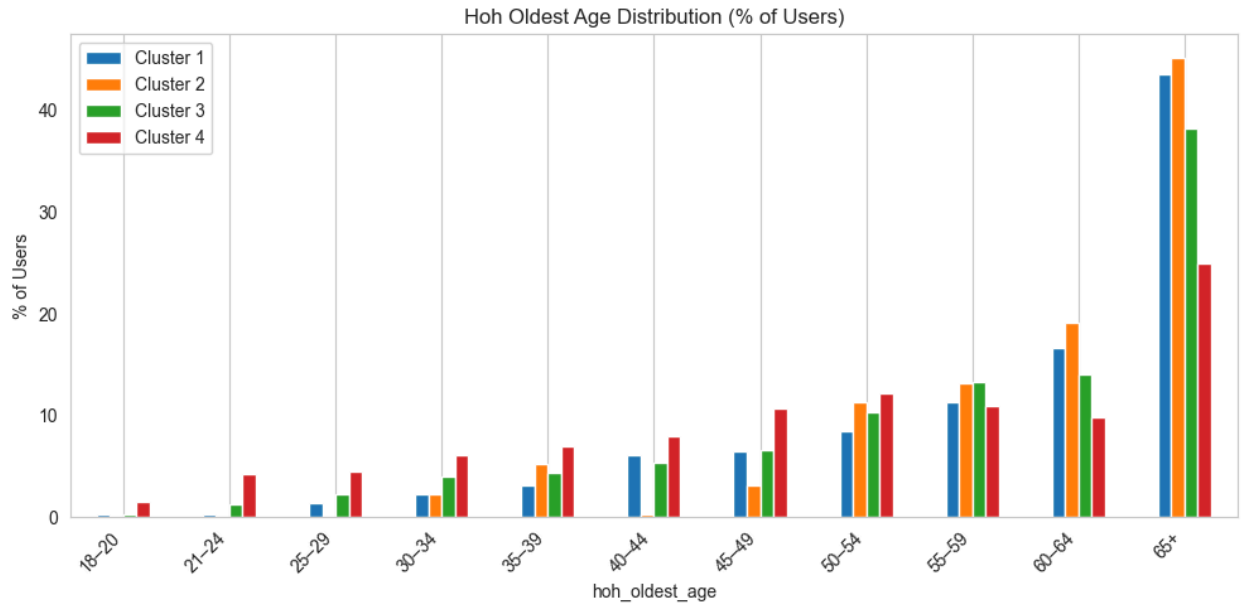
Based on the 4-means clustering, here are the number of people in each cluster.

Cluster	Number of people	Percentage
1	26,711	27.0%
2	48,115	48.6%
3	7,680	7.76%
4	16,395	16.6%

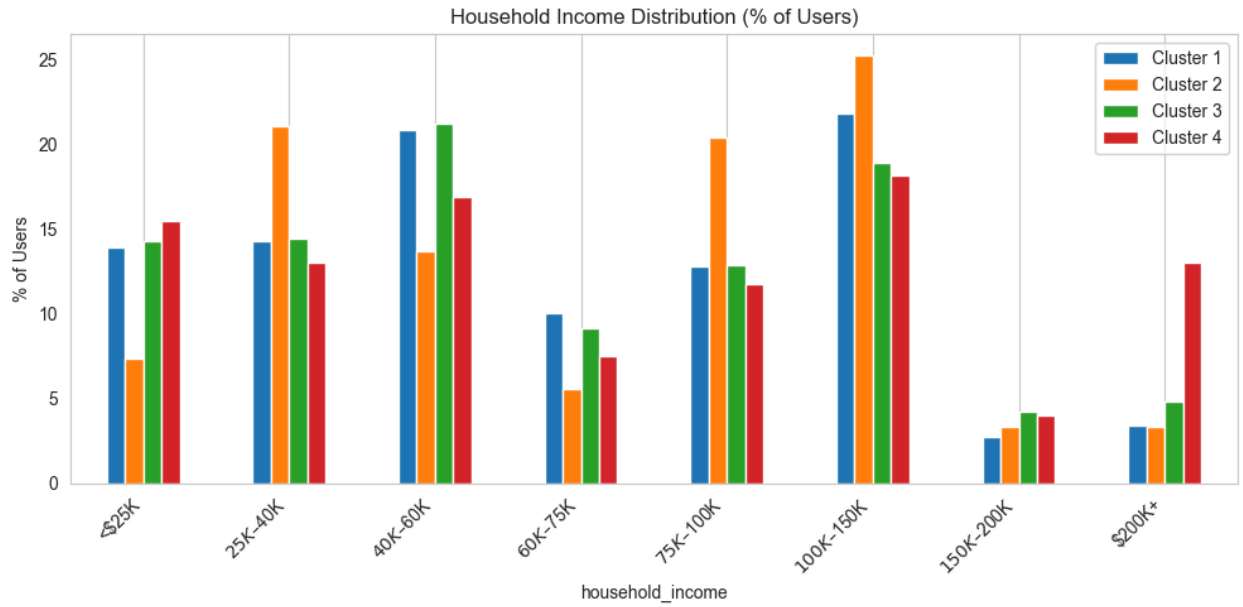
As we dig into the demographics of each cluster, we found interesting, different trends as follows:



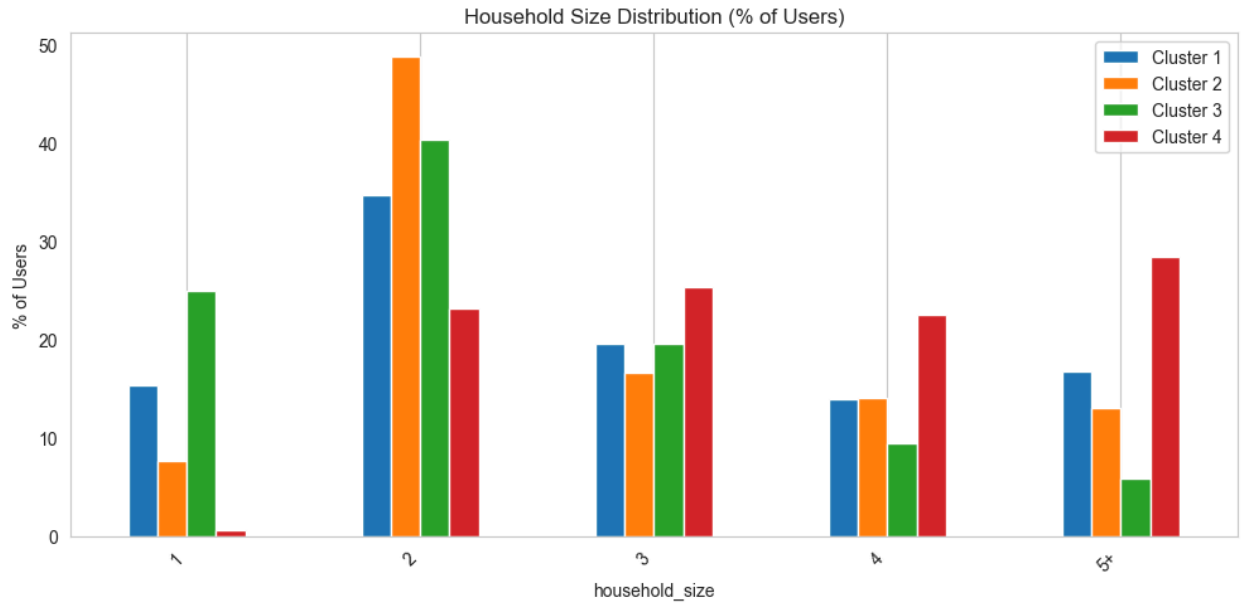
Racial background: Caucasian users dominate all clusters, with cluster 103 making for ~80% of users. However, Cluster 4 has a higher percentage of users with unknown backgrounds.



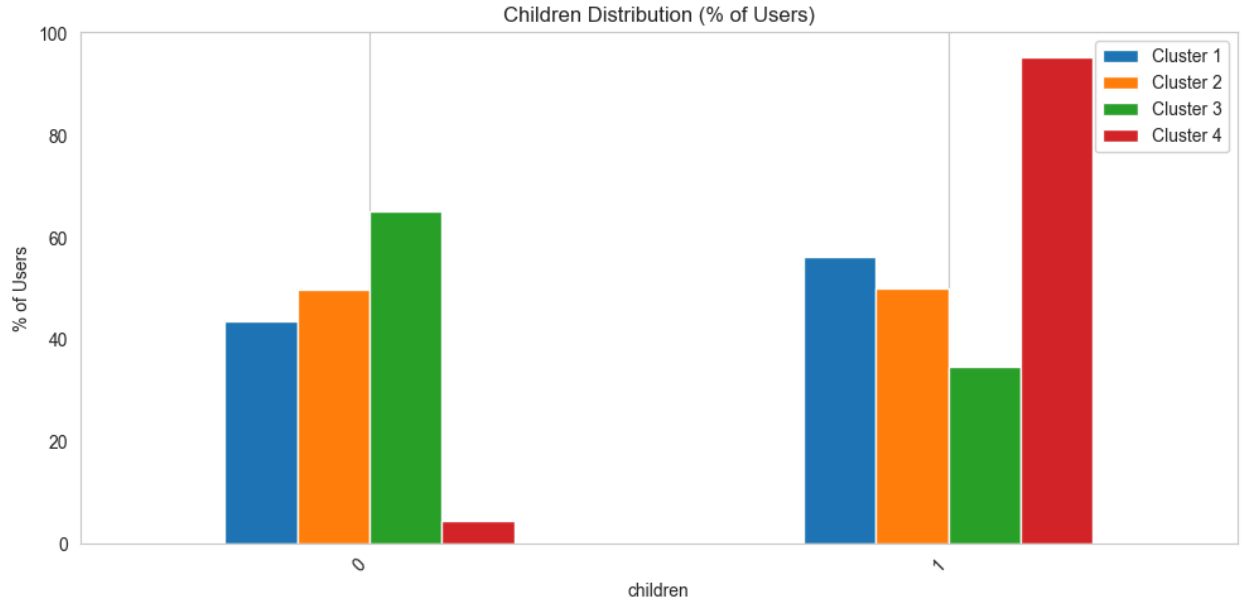
Household oldest age: The 65+ age group dominates all clusters. Cluster 2 skews with older people in the household, with 80-90% of people aged 50+. In contrast, Cluster 4 skews younger, with relatively higher proportions in the 18–44 age brackets compared to other clusters.



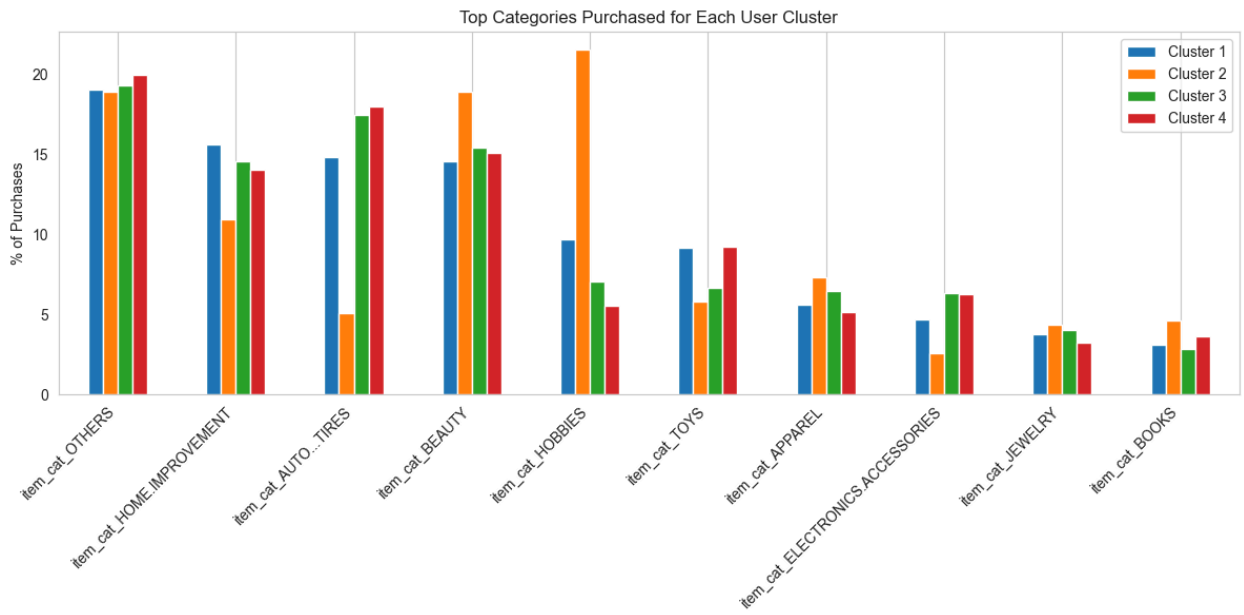
Household income distribution: Cluster 2 is peaking around mid-to-upper incomes, peaking at $100k-$150k. Cluster 4 has a more even spread, but with a stronger presence at both low (<$25k) and high (>$200k) ends. Clusters 1 and 3 lean toward the middle class, with $40k–$150k being dominant.



Household size distribution: Cluster 2 is dominated by 2-person households (49%), suggesting smaller, likely dual-income or couple households. Cluster 4 trends toward larger households (3 to 5+), with the highest representation in 3+ person homes. Cluster 3 has a balanced mix but skews slightly toward 2-person homes. Lastly, Cluster 1 appears relatively spread out, with notable presence in both 1-person and 2-person households.



Having children: While cluster 1 and 2 are pretty balanced on having or not having children, cluster 3's majority (>60%) doesn't have children, and **cluster 4's majority (95%) has children.**



Top categories purchased:

- **Cluster 1** appears balanced across HOME_IMPROVEMENT, AUTO_TIRES, and BEAUTY, with moderate representation in most categories.
- **Cluster 2** heavily over-indexes on HOBBIES (22%) and BEAUTY.
- **Cluster 3** prefers ELECTRONICS_ACCESSORIES and JEWELRY more evenly than others, aligning with self-use or giftable items.
- **Cluster 4** shows stronger interest in AUTO_TIRES, TOYS, and HOME_IMPROVEMENT, suggesting family-oriented, practical purchases.

Logistic Regression Results

1. Cluster 1

```
Coefficients:
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                     -1.830e+00  7.044e-02 -25.977  < 2e-16 ***
pages_viewed                     9.985e-03  3.773e-04  26.465  < 2e-16 ***
household_size                  -5.946e-02  6.180e-03  -9.623  < 2e-16 ***
hoh_oldest_age                   5.189e-02  3.883e-03  13.364  < 2e-16 ***
household_income                -1.780e-02  4.207e-03  -4.231 2.33e-05 ***
hispanic                        -2.235e-01  2.762e-02  -8.092 5.86e-16 ***
item_cat_OTHERS                  2.137e+01  1.362e+02   0.157 0.875354
item_cat_BEAUTY                  2.147e+01  1.637e+02   0.131 0.895607
item_cat_HOME.IMPROVEMENT        2.150e+01  1.580e+02   0.136 0.891801
item_cat_AUTO...TIRES            2.134e+01  1.589e+02   0.134 0.893173
item_cat_HOBBIES                 2.045e+01  1.501e+02   0.136 0.891689
item_cat_TOYS                    2.082e+01  1.697e+02   0.123 0.902349
item_cat_APPAREL                 2.124e+01  2.587e+02   0.082 0.934564
item_cat_ELECTRONICS.ACCESSORIES 2.157e+01  3.032e+02   0.071 0.943303
item_cat_JEWELRY                 2.114e+01  2.952e+02   0.072 0.942902
item_cat_BOOKS                   2.103e+01  3.091e+02   0.068 0.945763
race_Caucasian                   2.676e-01  2.673e-02  10.010  < 2e-16 ***
race_Other                       3.470e-01  3.884e-02   8.934  < 2e-16 ***
region_North_East               -1.019e-01  2.038e-02  -5.001 5.69e-07 ***
region_North_Central             7.139e-02  2.007e-02   3.557 0.000375 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 228323  on 164699  degrees of freedom
Residual deviance: 100175  on 164680  degrees of freedom
AIC: 100215

Number of Fisher Scoring iterations: 20
```

There are 9 statistically significant features:

1. Pages_viewed
2. Household_size
3. Hoh_oldest_age
4. Household_income
5. Hispanic
6. race_Caucasian
7. race_Other
8. region_North_East
9. region_North_Central

Feature interpretations:

Feature	Coef (β)	Odds Ratio (e^β)	% Change in Odds	Interpretation
pages_viewed	0.009985	1.0100	+1.0%	Each additional page viewed increases the odds by ~1%.
household_size	-0.05946	0.9423	−5.8%	Each additional person in the household decreases the odds by ~5.8%.
hoh_oldest_age	0.05189	1.0532	+5.3%	Each age group increase raises the odds by ~5.3%.
household_income	-0.01780	0.9824	−1.8%	Each income level increase slightly decreases the odds.
hispanic	-0.2235	0.7997	−20.0%	Hispanic users are ~20% less likely to convert.
race_Caucasian	0.2676	1.3067	+30.7%	Caucasian users have ~30.7% higher odds.
race_Other	0.3470	1.4146	+41.5%	"Other" racial groups have ~41.5% higher odds.
region_North_East	-0.1019	0.9032	−9.7%	Users in the Northeast are ~9.7% less likely to convert.
region_North_Central	0.07139	1.0741	+7.4%	North Central users are ~7.4% more likely to convert.

Combining with the user analysis from the graph visualizations, we get the following user persona for Cluster 1: homeowners/couples investing in their living space and self-care

We can see that this customer cluster is a focused, efficient group who knows what they are looking for. They don't live in a large household but make decisions quickly. Demographics and regional factors play a notable role in shaping their likelihood to take action (e.g., make a purchase, convert, etc.).

This group primarily consists of users from mid-sized households (typically 2–3 members), with moderate income levels and largely Caucasian backgrounds. Their demographic makeup suggests a stable, traditional user base, likely couples or small families managing household needs.

How They Behave
These users show strong interest in practical categories like **home improvement**, **auto & tires**, and **beauty**. Their browsing behavior is steady but not excessive, indicating a purposeful and efficient shopping style. They aren't impulsive but they do convert consistently, reflecting deliberate and value-oriented decision-making.

What Drives Them
This segment appears motivated by functionality and personal upkeep. Their purchasing patterns suggest they may be homeowners or couples investing in their living space, self-care, or vehicle maintenance.

Business Insight
This group forms a stable foundation of eBay shoppers. Marketing efforts that highlight upgrades, seasonal DIY projects, or value bundles in practical categories are likely to resonate.

2. Cluster 2

```
Coefficients: (1 not defined because of singularities)
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                     -1.367e+00  8.652e-02 -15.798  < 2e-16 ***
pages_viewed                     5.943e-03  4.845e-04  12.266  < 2e-16 ***
household_size                   8.658e-03  1.070e-02   0.809 0.418505
hoh_oldest_age                   4.146e-02  7.180e-03   5.775 7.71e-09 ***
children                         2.567e-01  2.570e-02   9.988  < 2e-16 ***
item_cat_BEAUTY                  2.062e+01  2.113e+02   0.098 0.922266
item_cat_HOME.IMPROVEMENT        2.098e+01  3.227e+02   0.065 0.948163
item_cat_AUTO...TIRES            2.097e+01  4.738e+02   0.044 0.964695
item_cat_HOBBIES                 1.929e+01  1.351e+02   0.143 0.886420
item_cat_TOYS                    2.042e+01  4.011e+02   0.051 0.959406
item_cat_APPAREL                 1.996e+01  2.958e+02   0.067 0.946197
item_cat_ELECTRONICS.ACCESSORIES 2.158e+01  7.800e+02   0.028 0.977932
item_cat_JEWELRY                 2.051e+01  4.045e+02   0.051 0.959564
item_cat_BOOKS                   1.904e+01  2.750e+02   0.069 0.944818
race_Caucasian                  -2.493e-01  3.015e-02  -8.267  < 2e-16 ***
region_South                    -1.573e-01  2.795e-02  -5.628 1.83e-08 ***
connection_speed                        NA         NA      NA       NA
region_North_Central             1.150e-01  3.482e-02   3.301 0.000962 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 68175  on 49177  degrees of freedom
Residual deviance: 40142  on 49161  degrees of freedom
AIC: 40176

Number of Fisher Scoring iterations: 20
```

There are 6 statistically significant features:

1. Pages_viewed
2. hoh_oldest_age
3. Children
4. race_Caucasian
5. gegion_South
6. region_North_Central

Feature interpretation:

Feature	Coef (β)	Odds Ratio	% Change in Odds	Interpretation
pages_viewed	0.005943	1.006	+0.6% per page	Each page viewed slightly raises the odds of converting.
hoh_oldest_age	0.04146	1.0423	+4.2% per age group	Having older people in the household makes them more likely to convert.
children	0.2567	1.2926	+29.3%	Households with children are significantly more likely to convert.
race_Caucasian	−0.2493	0.7793	−22.1%	Caucasian users are less likely to convert.
region_South	−0.1573	0.8545	−14.6%	Users in the South are less likely to convert.
region_North_Central	0.1150	1.1219	+12.2%	North Central users are more likely to convert.

Based on the results and the visualizations, we believe that Cluster 2 is a middle-aged woman who looks to upgrade their lifestyles after work/retirement.

Cluster 2 users typically come from smaller households (mostly 2 people, could be their partner or their child) and span a wider income range. Most of the people in this group have at least one middle-to-old person (age 50+) in the household.

How They Behave

They are highly engaged browsers, especially within **hobbies** and **beauty** categories. They explore more before buying, with the lowest page view conversion rate of 0.6%. They tend to make passion- or interest-driven purchases. Having a child can help them buy more products, as they could be doing activities with their child.

What Drives Them

This group seems to be guided by personal interests or lifestyle upgrades. Think solo crafters, a retired or nearly-retired woman, or a stay-at-home mother. Their behavior is shaped more by what captures their attention than by household needs.

Business Insight

These are great targets for inspiration-based marketing. Personalized recommendations,

product discovery features, and interest-based bundles (like hobby kits or new arrivals in beauty) could drive further engagement.

3. Cluster 3

```
Coefficients:
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                     -2.153e+00  6.939e-02 -31.034  < 2e-16 ***
pages_viewed                     1.966e-02  5.175e-04  37.993  < 2e-16 ***
household_size                   3.770e-02  8.311e-03   4.536 5.74e-06 ***
hoh_oldest_age                   4.494e-02  3.730e-03  12.050  < 2e-16 ***
household_income                -1.319e-02  4.273e-03  -3.085  0.00203 **
children                         5.774e-01  1.883e-02  30.655  < 2e-16 ***
item_cat_OTHERS                  2.064e+01  1.010e+02   0.204  0.83801
item_cat_BEAUTY                  2.054e+01  1.117e+02   0.184  0.85409
item_cat_HOME.IMPROVEMENT        2.080e+01  1.221e+02   0.170  0.86467
item_cat_AUTO...TIRES            2.068e+01  1.067e+02   0.194  0.84634
item_cat_HOBBIES                 1.970e+01  1.248e+02   0.158  0.87460
item_cat_TOYS                    2.017e+01  1.533e+02   0.132  0.89538
item_cat_APPAREL                 2.033e+01  1.661e+02   0.122  0.90257
item_cat_ELECTRONICS.ACCESSORIES 2.058e+01  1.787e+02   0.115  0.90834
item_cat_JEWELRY                 2.017e+01  1.961e+02   0.103  0.91808
item_cat_BOOKS                   2.029e+01  2.408e+02   0.084  0.93284
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 196355  on 141639  degrees of freedom
Residual deviance:  85174  on 141624  degrees of freedom
AIC: 85206

Number of Fisher Scoring iterations: 19
```

Only 5 features are statistically significant:

1. Pages_viewed
2. Household_size
3. Hoh_oldest_age
4. Household_income
5. Children

Feature interpretations:

Feature	Coef (β)	Odds Ratio	% Change in Odds	Interpretation
pages_viewed	~0.0198	~1.020	+2.0%	Each additional page viewed increases the odds of the outcome by ~2.0%.
household_size	~0.0373	~1.038	+3.8%	Each additional person in the household increases the odds by ~3.8%.
hoh_oldest_age	~0.0450	~1.046	+4.6%	Each unit increase in age group increases the odds by ~4.6%.
household_income	~-0.0132	~0.987	−1.3%	Each unit increase in income reduces the odds by ~1.3%.
children	~0.5774	~1.781	+78.1%	Having children in the household increases the odds by ~78.1%.

Based on the results and the visualizations, we believe that Cluster 3 is a value-conscious, stable, tradition family.

These users tend to live in moderate to large households and often don't have children. Their household size is more skewed towards living alone or with another person. They skew slightly older and more budget-conscious, with a demographic leaning toward middle-to-lower income groups, with a peak at $40-60k. The majority are Caucasian, though the cluster includes a broader spread. Similar to Cluster 2, these people are living with a middle to old person (age 50+) in their household.

How They Behave
They browse frequently and are responsive to both household size (the bigger the household, the more likely they are purchasing), and the presence of children (having children strongly pushes them to purchase). Their purchasing decisions are shaped by budget considerations and the frequency of their page visits. The top categories they are purchasing are mostly household products such as **auto and tires, beauty products, and home improvement** products.

What Drives Them
This cluster is driven by family needs and financial value. They prioritize practicality in shopping, cost-efficiency, thoughtful comparison, and timing purchases carefully.

Business Insight

This is the value-conscious family segment. Promotions that focus on savings, multipacks, or family-friendly bundles, especially in relevant categories will help convert these users. Retargeting strategies that reward returning browsers can help convert them.

4. Cluster 4

```
Coefficients:
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                      -3.002343   0.142815 -21.023  < 2e-16 ***
pages_viewed                      0.028881   0.001419  20.359  < 2e-16 ***
household_size                    0.212252   0.017629  12.040  < 2e-16 ***
children                          0.746631   0.125451   5.952 2.66e-09 ***
item_cat_OTHERS                  20.942503 309.193832   0.068    0.946
item_cat_BEAUTY                  20.818018 343.689575   0.061    0.952
item_cat_HOME.IMPROVEMENT        21.015760 367.847069   0.057    0.954
item_cat_AUTO...TIRES            20.893194 318.339997   0.066    0.948
item_cat_HOBBIES                 20.217556 463.683549   0.044    0.965
item_cat_TOYS                    20.787315 447.361270   0.046    0.963
item_cat_APPAREL                 20.617904 552.913013   0.037    0.970
item_cat_ELECTRONICS.ACCESSORIES 20.841941 561.387080   0.037    0.970
item_cat_JEWELRY                 20.489381 675.841419   0.030    0.976
item_cat_BOOKS                   20.661654 662.071885   0.031    0.975
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 32780  on 23645  degrees of freedom
Residual deviance: 15855  on 23632  degrees of freedom
AIC: 15883

Number of Fisher Scoring iterations: 19
```

There are 3 statistically significant features:

1. Pages_viewed
2. household_size
3. Children.

Feature interpretations:

Feature	Coef (β)	Odds Ratio	% Change in Odds	Interpretation
pages_viewed	~0.0289	~1.0293	+2.93%	Each additional page viewed increases the odds of the outcome by ~2.93%.

household_size	~0.2122	~1.236	+23.6%	Each additional person in the household increases the odds by ~23.6%.
children	~0.7466	~2.110	+111%	Having children in the household more than doubles the odds of the outcome (~111% increase).

Based on the results and the visualizations, Cluster 4 is a unique group: a younger family with children and spread-out income.

Who They Are
This small but distinct group stands out due to their large households (mostly 4–5+ people) and high percentage of users with children. They also include more users from the highest and lowest ends of the income spectrum, and are younger on average. Racial backgrounds are more varied, and a greater proportion are marked as unknown.

How They Behave
These users are highly active, browsing many pages and frequently converting, especially when children and household size are factors. They don't stick to specific categories, showing interest across **auto & tires**, **beauty**, and a wide spread of "other" items. Their behavior is less predictable but more action-oriented once engaged.

What Drives Them
This group is strongly influenced by **family context**. With high household needs and possibly shared devices or accounts, their browsing may reflect group shopping behavior. The presence of children more than doubles the odds of purchase, making them highly reactive once a relevant need is triggered.

Business Insights
This cluster responds well to dynamic and varied offerings. Think cross-category deals, family-focused promotions, and multi-pack discounts. Their spontaneity makes them ideal targets for limited-time offers or triggered messages based on browsing patterns.